

Clinical Decision Support System - Proposed Extensions

Base Paper: An Adaptive Multi-Agent LLM-Based Clinical Decision Support System Integrating Biomedical RAG and Web Intelligence (IEEE Access, 2025)

Existing System:

1. Clinical Text Clarifier Agent
2. Symptom RAG Analyzer Agent
3. Evidence-Based Scanner Agent
4. Clinical Data Fusion Agent
5. Adaptive Optimizer Agent

Limitations:

- Text-only diagnosis
- No medical image analysis
- No risk prediction
- No specialist recommendation
- No treatment pathway generation
- Limited explainability

Proposed Extensions

1. Medical Image Analysis Agent

Existing: Uses only symptoms and clinical text.

Extension: Analyze X-rays, CT scans, and MRI images and combine findings with symptom-based diagnosis.

2. Patient Risk Prediction Agent

Existing: Diagnosis only.

Extension: Predict Low/Medium/High patient risk using age, history, lab reports, and comorbidities.

3. Specialist Recommendation Agent

Existing: Diagnosis output only.

Extension: Recommend relevant specialists such as Pulmonologists, Cardiologists, etc.

4. Personalized Treatment Pathway Agent

Existing: No treatment planning.

Extension: Suggest tests, follow-up procedures, and treatment pathways based on diagnosis and risk level.

5. Explainable Diagnosis Dashboard

Existing: Confidence score only.

Extension: Provide reasons behind each diagnosis and visual explanation.

Final Architecture

- Clinical Text Clarifier Agent
- Symptom RAG Analyzer Agent
- Evidence-Based Scanner Agent
- Clinical Data Fusion Agent
- Adaptive Optimizer Agent
- Medical Image Analysis Agent
- Patient Risk Prediction Agent
- Specialist Recommendation Agent

- Personalized Treatment Pathway Agent
- Explainable Diagnosis Agent

Expected Contributions

1. Multimodal Clinical Decision Support
2. Patient Risk Prediction
3. Specialist Recommendation
4. Personalized Treatment Planning
5. Explainable AI for Healthcare